

as part of the repository. The data is categorised and organised so that it can be used by generative AI models.

Data monitoring: This step involves monitoring the quality and relevance of the data, and updating it to improve the performance of the LLM.

Data governance: This step covers the automation of policies, guidelines, principles, and standards. Data governance must align with AI governance at the enterprise level to realise business objectives.

LLM models can be built using open source or proprietary models. Open source models are off-the-shelf and customisable, while proprietary models are offered as LLMs-as-a-service. Figure 2 shows a few LLM tools.

Let's now take a look at the use cases of LLMs across industries.

Healthcare and pharma

Generative AI-based applications help healthcare professionals be more productive, identifying potential issues upfront, and providing insights to deliver interconnected health. This helps in:

Better customer experience:

Automating administrative tasks, such as processing claims, scheduling appointments, and managing medical records.

Patient health summary:

Providing healthcare decision support by generating personalised patient health summaries, thus improving patient response times and experience.

Faster analysis of publications:

Reducing the time it takes to create research publications on specific drugs by analysing vast amounts of data from multiple sources faster than ever. This helps in accelerating the speed and quality of care, and can also improve drug adherence.

Personalised medicine: Creating individual treatment plans based on a patient's genetic makeup, medical history, life style, etc.

Healthcare virtual assistant:

Providing end users with conversational and engaging access to the most relevant and accurate healthcare services and information.

Manufacturing

Generative AI enables manufacturers to create more with their data, leading to advancements in predictive maintenance and demand forecasting. It also helps in simulating manufacturing quality, improving production speed, and efficiency in the use of materials.

Predictive maintenance: Helps in estimating the life of machines and their components, proactively providing information to technicians about repairs and replacement of parts and machines. This helps in reducing the downtime.

Performance efficiency:

Anticipates production problems in real-time, covering the risks of production disruptions, bottlenecks, and safety.

Other usages of generative AI in the manufacturing industry include:

- Yield, energy, and throughput optimisation
- Digital simulations
- Sales and demand forecasting
- Logistics network optimisation

Input	Generative AI LLM Tools	Results
Text Generation & Chatbots Tools	Jasper Copy.ai OpenAI AYLIEN LinkDelta Curated AI Crystal Ryters Block Automated Insights ChatGPT Notion AI Artificial Fiesta agolo alltius Contlo InsurStaq.ai LONGSHOT Scalenut Social Beat senseforth.ai SYNCSENSE ZPOD	Blog posts, Articles, Product descriptions, Creative writing, Language Translation
Image Generation & 3D Modelling Tools	OpenAI Canva breeze.ai Wix.com FIREDROP Poly PRISMA Photo Room Booth.ai Adobe Sensei GRAPHIG BlendAI HEXO Nuro Pixel.AI Postify AI Spyne Uns Zapero Predis.ai Zocket Magic studio Pebblely	Image Generation, Image Captioning
Audio & Video Generation Tools	Fliki abridge amper Voicemaker Jukedeck audio bot Speechfy DeepMind Lyrebird Visla AI Topaz AI Synthesia HeyGen WIBBITZ 03 runway Scalenut SOLOSHOT WSC Sports WordAI Magesto Syllaby beatoven.ai dubpro.ai Listnr kroop AI MURF AI Predis.ai Rephrase.ai SYNCSENSE trueeach.ai Unscript.ai vidyo.ai Vitra.ai	Text to Speech, Instruction Following, Chatbot builder Video Generation, Visually captivating videos, Text to Video, Streamline video creation process
Code & Data Generation Tools	GitHub copilot Replit Ghostwriter Codemate eizen FluidAI hype ai kombai observe-AI pixis workhack portkey Maya DhiWise CodeGPT	Generate code snippets, Respond to queries, Assisting developers with writing code, debugging existing code bases

Figure 2: LLM tools